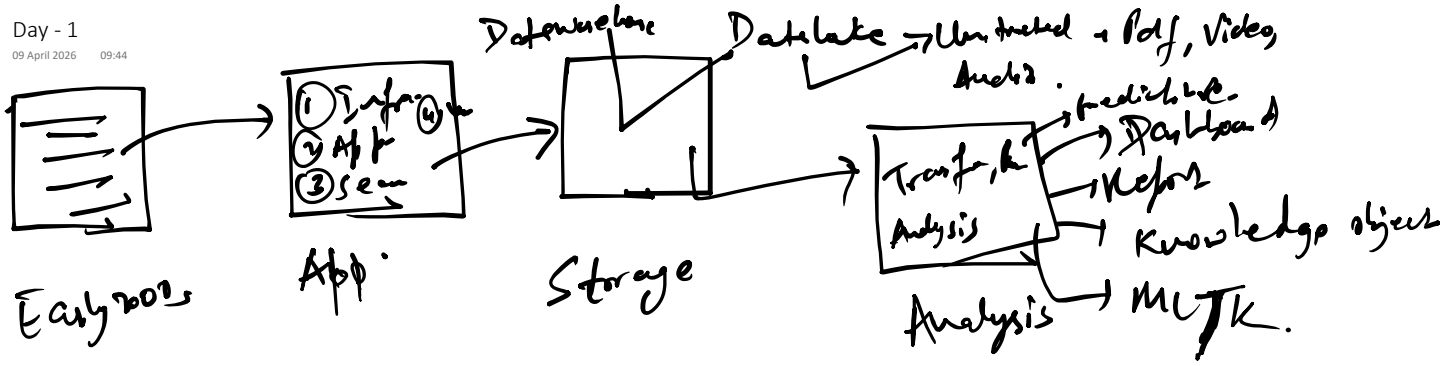
Day - 1

09 April 2026 09:44



Data warehouse → Structured — CSV

Semi structured — XML, json

Monitoring → Reactive Approach.

Observability → Proactive Approach.

3 Pillars of observability

1. Log
2. Metrics
3. Traces.

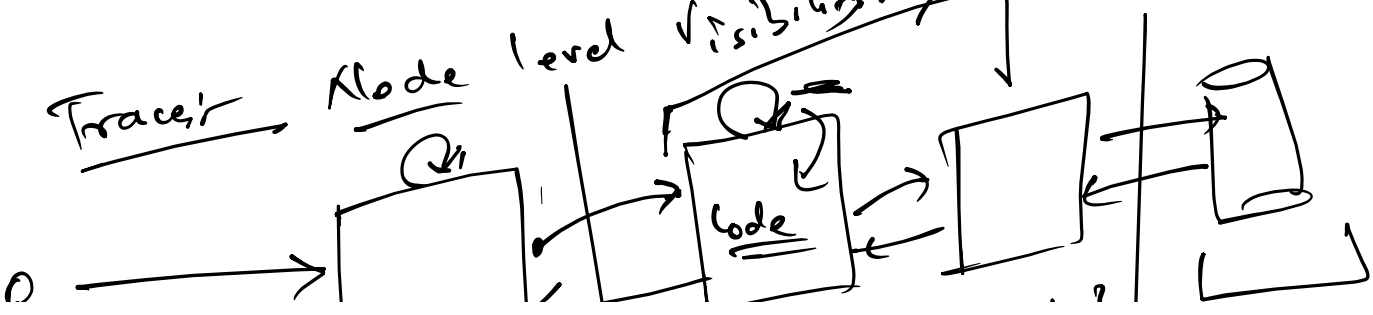
Metrics → Aggregate Value.

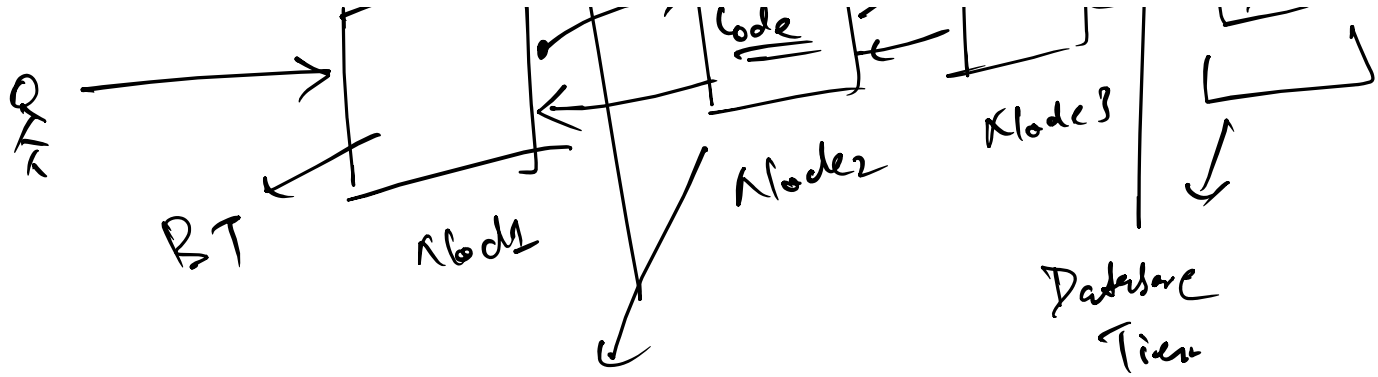
CPU = 80% → CPU = critical

Logs → Timestamp details of that Activity.

CPU = 80% ip = —

Traces → Node level visibility.



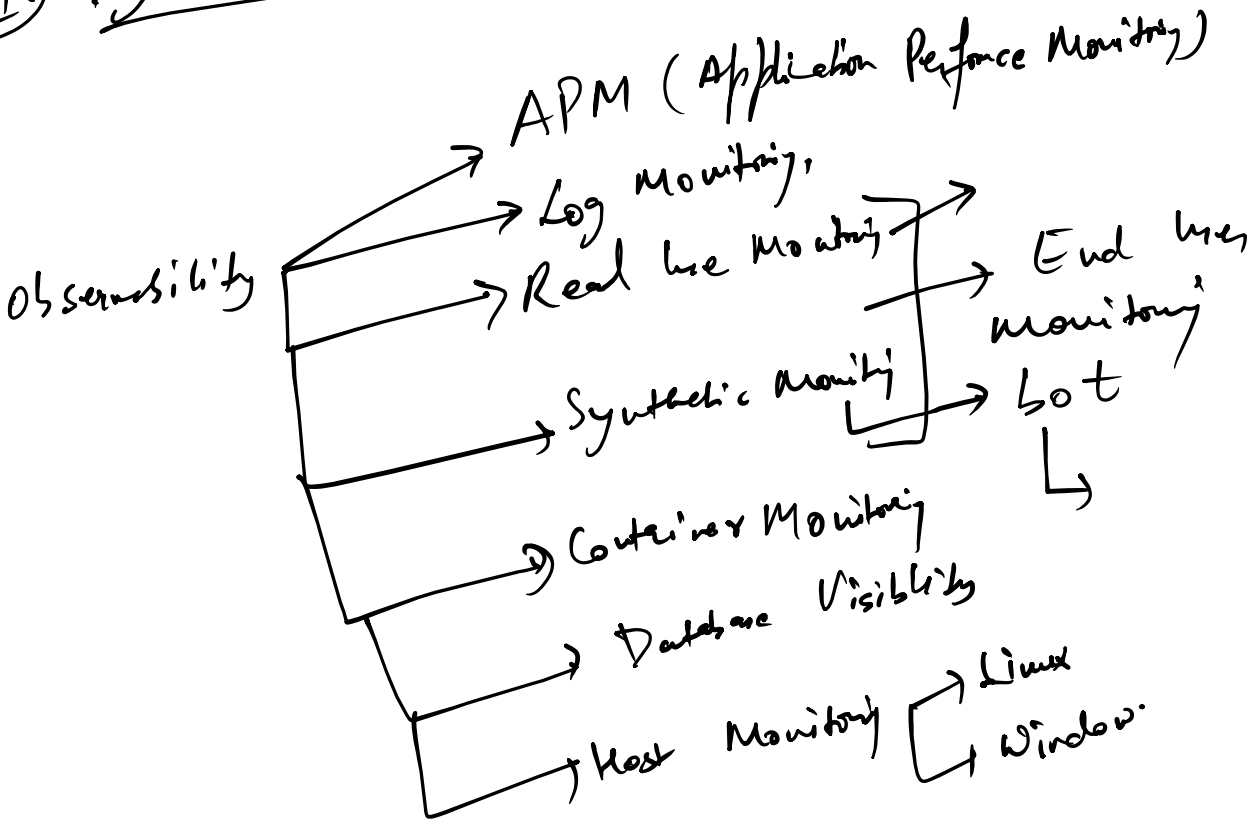


Metrics + Logs + Traces → Observability → Trend

Observability:-

1. Dynatrace.
2. New Relic
3. AppDynamic
4. Splunk
5. ELK
6. Datadog.

1. Dynatrace:-



Dynatrace – Observability Tool

Around 2014

DT → Appmon
DT → DCRUM – Datacentre RUM
DT → Enterprise Synthetic
DT → Gomez

Ruxit, → Dynatrace

DT → SaaS → URL
SaaS → Software As a Service

DT → Managed → Infra + App
Infra + App → cloud (AWS, Azure)
Infra + App → On-prem

Hybrid

① Dynatrace Saas Architecture:-

OneAgent → 443 → DT Server
OneAgent → 443 → DT Server
OneAgent → 443 → DT Server

① Application – Java / .Net.
② Infra. – Linux / Windows

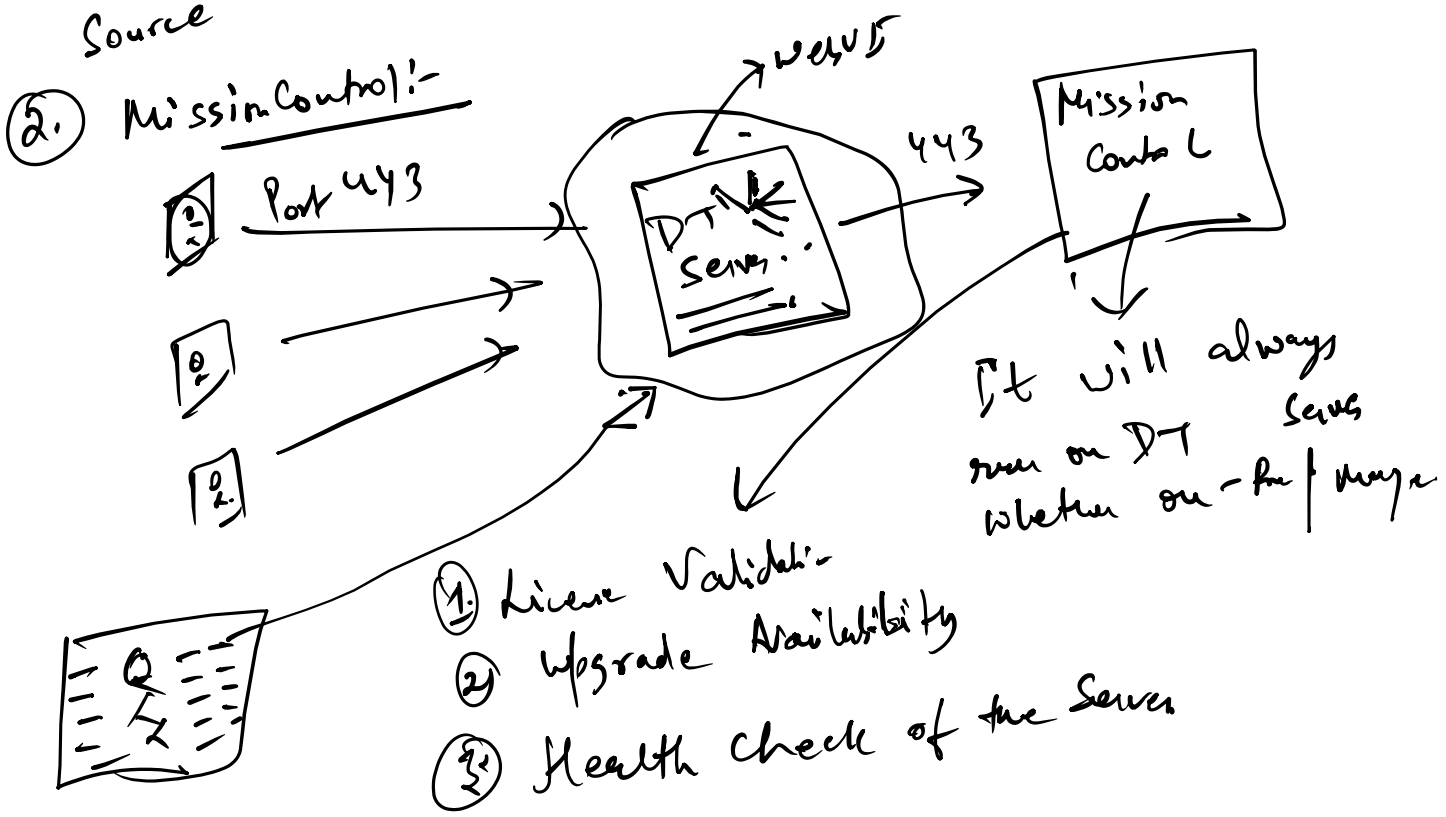
AppD → extension → .Net / JS / Java

OneAgent is dependent on OS not on App Type.

One Agent is deployed on OS not on App Type.

Source

2. MissionControl:-

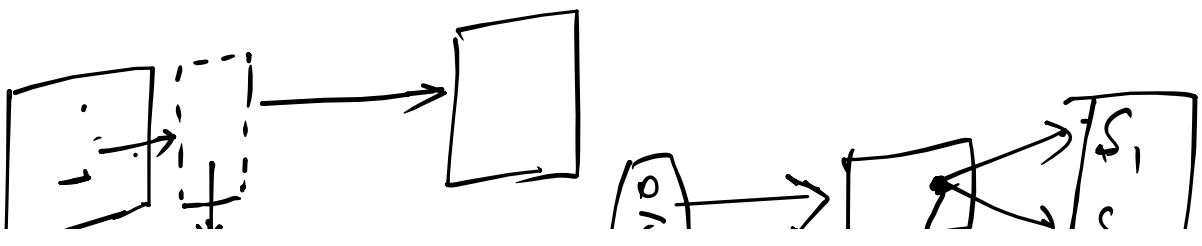


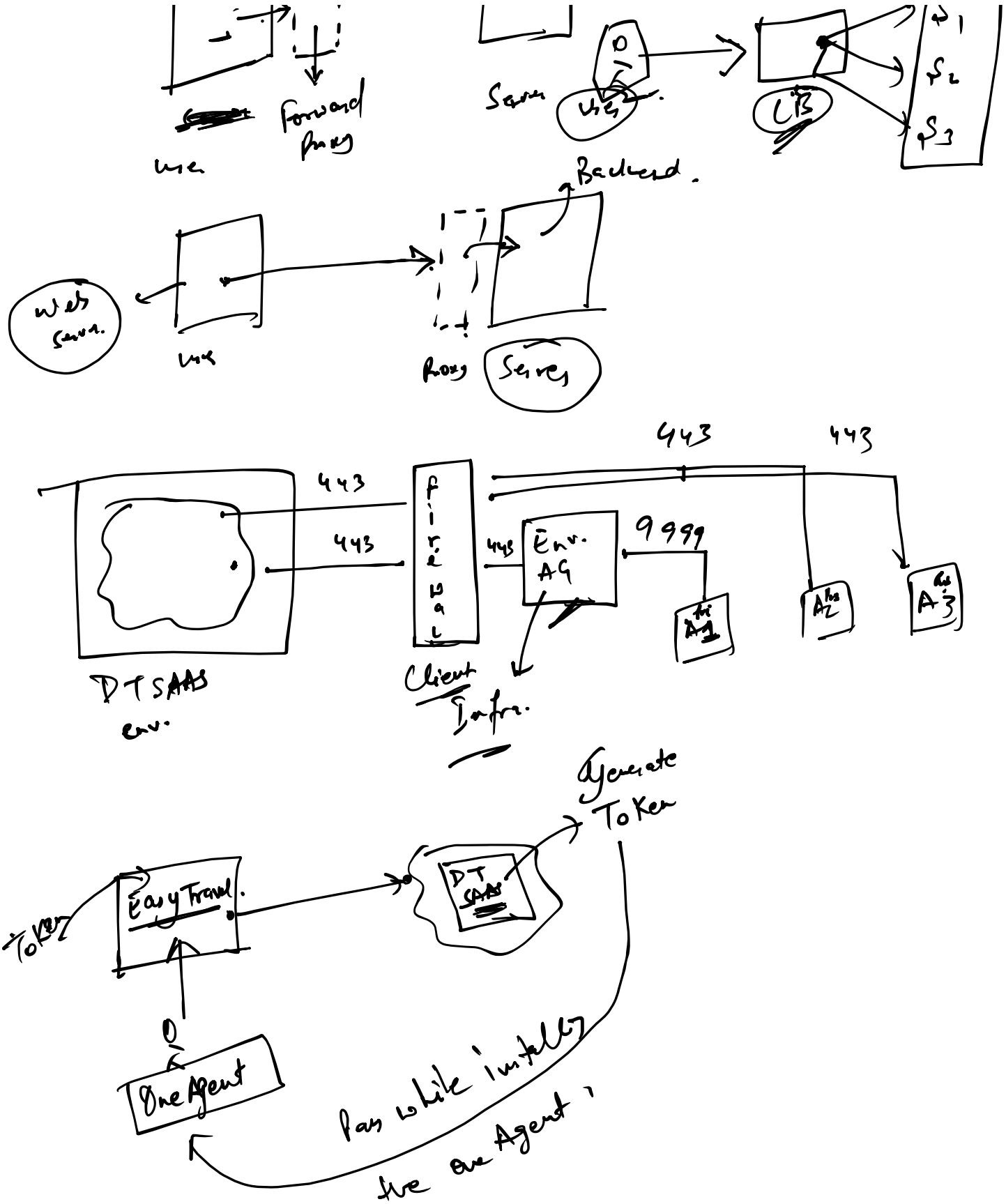
1. License Validation
2. Upgrade Availability
3. Health check of the Server

DT Server Components:-

1. Nginix → Reverse Proxy Server
2. Elastic Search → Analyse the data.
3. Cassendra HyberCube. → Distributed DT
4. DT Server →

Proxy Server
- Forward Proxy Server
- Reverse Proxy Server.





(1) API Token. (Rest API Access)

(2) Paas Token (Core Agent / Active gate)

2. Paas Token Core Mgmt

Scope = Permission Assigned to a token.

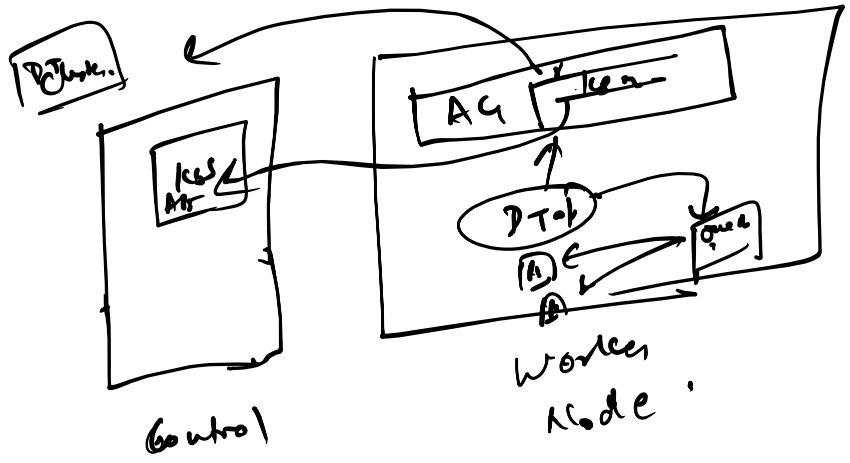
Why!

1. Security
2. Controlled Automation
3. Reduce blast radius.
   - extent of damage
4. Prevent misuse.

Monitoring :-

- → Fullstack – end to end visibility
  - APM
  - security AAI
- → Infra – Host
  - VMs
  - Container
  - N/W & storage.
- → Discovery
  - → Service
    - Process
    - Host
    - Dependence

1. Dynatrace operation.
2. Data ingest – logs | Metrics | Event | Infra.



Control Plane

Worker Node

Reachability — Network Connective status.

- Reach other services?
- Reach this host? → Ubuntu.

✓ → N/w conn. is working/fine

X → Not reachable → firewall, port

– → Not enough data / not applicable

Page faults — Process tries to Access memory that is not currently in RAM

- Minor Page fault (Soft faults)
- Major Page fault (Hard faults)

App (data) → not in RAM → OS load from Disk → Major Page fault

App (data) → Already in RAM → just mapped → Minor page fault.

Higher Page fault ↑ Perform ↓

Swap Usage — Disk space used as extra. RAM

RAM full → move old data → swap

need data again → bring back → page fault occurs

Disk is much slower than RAM.

Heavy swap usage ↑ system slow ↓

Low RAM ∝ OS uses swap.

RAM full → Data moved to swap → App need it → Disk read → page fault → slow